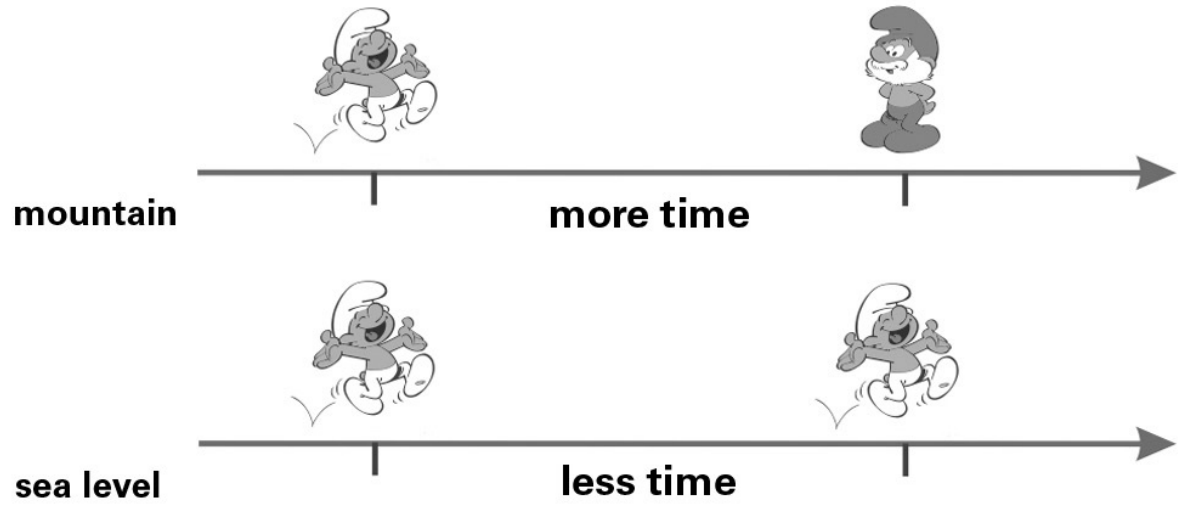


Is this surprising? Perhaps it is. But this is how the world works. Time passes more slowly in some places, more rapidly in others.

The surprising thing, perhaps, is that someone understood this slowing down of time a century before we had clocks precise enough to measure it. His name, of course, was Albert Einstein.

The ability to understand something before it's observed is at the heart of scientific thinking. In antiquity, Anaximander understood that the sky continues beneath our feet long before ships had circumnavigated the Earth. At the beginning of the modern era, Copernicus understood that the Earth turns long before astronauts had seen it do so from the moon. In a similar way, Einstein understood that time does not pass uniformly everywhere *before* the development of clocks accurate enough to measure the different speeds at which it passes.

In the course of making such strides, we learn that the things that seemed self-evident to us were really no more than prejudices. It seemed obvious that the sky was above us and not below; otherwise, the Earth would fall down. It seemed self-evident that the Earth did not move; otherwise, it would cause everything to crash. That time passed at the same speed everywhere seemed equally *obvious* to us. . . . Children grow up and discover that the world is not as it seemed from within the four walls of their homes. Humankind as a whole does the same.

Einstein asked himself a question that has perhaps puzzled many of us when studying the force of gravity: how can the sun and the Earth "attract" each other without touching and without utilizing anything between them?

He looked for a plausible explanation and found one by imagining that the sun and the Earth do not attract each other directly but that each of the two gradually acts on that which is between them. And since what lies between